

bring a falling elevator to a standstill if the cables broke. In 1861, Otis patented his steam engine for elevators but died the same year.

### **RAUSING, RUBEN (1895–1983)**

At lunch one day at home in Sweden, Rausing's wife Elizabeth suggested he invent something lightweight to transport liquids such as milk or juice. Head of a struggling food packaging company, Ruben rose to the challenge. In 1944, he patented the sterile, tetrahedron-shaped, plastic-coated carton he developed with Swedish engineer Erik Wallenberg. The company moved to the square design a few years later. Today, Tetra Pak is one of the most successful packaging companies in the world.

### **RITTY, JAMES (1836–1918)**

The cash register was the cunning idea of Ritty, the proprietor of an American bar. He came up against the problem of his takings disappearing without any record. He got together with his mechanic brother John and invented a machine with keys that could be pressed to record the amount of money taken at each transaction. In 1879, he patented his machine "Ritty's Incorruptible Cashier" and opened a factory in Dayton, Ohio, to produce his invention.

### **TALBOT, WILLIAM HENRY FOX (1800–1877)**

In 1835, Talbot made a photographic negative of the window of his home, Lacock Abbey, UK—it is the earliest surviving negative in the world. Talbot's love of drawing inspired him to find a different way of capturing images on paper. In 1841, he

perfected his technique of creating a calotype (early photograph) using a camera and paper treated with silver nitrate and potassium iodide.

### **TOYODA, SAKICHI (1867–1930)**

The Japanese industrialist and founder of the group that would become Toyota was also the inventor of the automatic power loom. Toyoda was 18 years old when he decided to become an inventor. In 1891, he patented his first invention—the wooden handloom, which required just one hand rather than two. By 1896, he had developed Japan's first steam-powered loom. Throughout his life, Toyoda made improvements to his original design, with patents granted all over the world.

### **SIKORSKY, IGOR (1889–1972)**

The Russian-American inventor of the helicopter had an early fascination with flight. He believed the best way to fly was to go straight up with a horizontal rotor. In Russia, Sikorsky came up with early unsuccessful prototypes for the helicopter. In 1919, he immigrated to the US, formed an aircraft company, and developed the first four-engine plane. Then, in 1939, he developed the first helicopter, the VS-300, which was used by the US military.

### **WAKEFIELD, RUTH GRAVES (1903–1977)**

Many inventions are a happy accident, and that is what happened when Wakefield baked cookies for the customers at her Massachusetts roadside inn around 1938. Wakefield claimed the chocolate chip cookies she created were like that by design, but other stories suggest that she

expected the broken bits of chocolate in her cookie mix to melt. Whichever story is true, Wakefield's "Toll House Chocolate Crunch Cookie" is the basis of America's favorite cookie.

### **WELLESLEY, ARTHUR (1769–1852)**

The famous Wellington boot takes its name from the British statesman and military leader Arthur Wellesley, the 1st Duke of Wellington. The war hero was often seen wearing his favorite leather boots. In the early 1800s, he was given a pair of Hessian boots by a group of German soldiers. Wellington asked his bootmaker to copy the design, but remove the tassel and extend the front over the knee for added protection. The style was copied by aristocrats everywhere and nicknamed the "Wellington." The rubber version was patented in 1852.

### **WOODS, GRANVILLE (1856–1910)**

Woods was an African American engineer who was granted more than 50 patents in his lifetime, many for inventions applying to the safety and improvement of the railroads. In 1887, he invented the Synchronous Multiplex Railway Telegraph, which enabled contact between a moving train and a station. Thomas Edison took him to court, claiming he had invented it first, but Woods won the case. He also helped improve inventions like the safety circuit, telephone, and phonograph, and invented an egg incubator and an automatic brake.

